

1. Product Architecture Overview

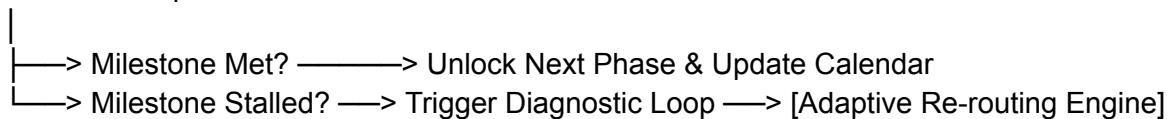
LaunchPad is an interactive, web-based decision-support platform designed to help creators, students, and early professionals clear cognitive overload, stress-test business or project hypotheses, and systematically execute tasks through a dynamic feedback loop.

[CONVERSATIONAL ENGINE]

Stage 1: Raw Idea Intake —> Stage 2: Assumption Interrogation —> Stage 3: Constraint Mapping



[INTERACTIVE DASHBOARD] <— Oauth Sync <— [GOOGLE CALENDAR] <— Stage 4 & 5: Core Roadmap



2. Deep-Dive Stage-by-Stage Functional Process

Stage 1: Idea Intake

- **User Experience:** A clean, minimal text interface with a single prompt: *"Tell me what you're trying to build or do."* No distracting options.
- **System Action:** Captures raw, conversational, unstructured text string.

Stage 2: Assumption Interrogation

- **System Action:** An LLM prompt chain parses the raw text against a standard business validation model (Target Customer $\rightarrow$ Value Proposition $\rightarrow$ Willingness to Pay $\rightarrow$ Delivery). It isolates the weakest logical link.
- **Example Output:** *"You mentioned building an on-demand laundry app for off-campus students. You are implicitly assuming students value time saving over the cost of doing it themselves. What initial evidence do you have for this?"*

Stage 3: Constraint Mapping

- **User Experience:** The user answers the interrogation questions. The platform follows up with a structural form-less question regarding constraints.
- **System Action:** Evaluates user assets: Total hours available per week, available budget, technical skill level, and "The Kill Switch" (the user's self-identified fatal flaw).

Stage 4: Structured Output Generation

The AI passes the accumulated context into a structured synthesizer template to output:

- **Idea Sharpening:** A concise, stripped-down summary of the project.
- **Top 3 Core Assumptions:** The foundational items that must be true for this project to survive.

- **30/60/90-Day Milestone Roadmap:** Milestones bound to hard durations (e.g., Milestone 1: 5 days, Milestone 2: 14 days).
- **The 48-Hour Micro-Action:** One simple, un-intimidating task to break initial inertia.
- **The Uncomfortable Risk:** A brutal, objective assessment of what could go wrong.

Stage 5: Human Decision Moment (Responsible AI Layer)

- **System Action:** A blockquote section renders explicitly: "This roadmap is a dynamic calculation, not an absolute truth. Milestone 1 requires deep subjective human intuition regarding user empathy, which this AI cannot simulate. You must decide whether these conditions make sense to execute."

3. Post-Agreement: Core Engine & Automation Integrations

The Google Calendar Sync Loop

1. **OAuth Consent:** Once the user clicks *"Accept & Commit to Roadmap"*, the web app requests standard Google OAuth scopes for `calendar.app.created` or `calendar.events`.
2. **Dynamic Generation:** LaunchPad creates a dedicated calendar named **"LaunchPad: [Project Name]"** to protect personal data layout hierarchy.
3. **Event Insertion:** The app translates the phased roadmap days into calendar block boundaries relative to `DateTime.now()`.
 - *Milestone 1 (Duration: 5 days):* Placed on Day 5.
 - *Proactive Check-In Event:* Automatically scheduled on Day 3 as a subtle, ambient notification reminder.

Gated Dashboard & Milestone Logic

The web dashboard is an interactive kanban or list timeline.

- **State Control:** Milestones 2 and 3 are explicitly rendered in a disabled, greyed-out visual state (`disabled=True`).
- **The Proactive Intercept (Day 2/3):** The application triggers an internal webhook. When the user logs onto the dashboard on Day 3, a personalized chat intercept pops open: *"Hey, you're 3 days into your 5-day goal to speak to 10 potential customers. How is the data looking so far?"*

4. The Adaptive Re-Routing Engine (Failure & Pivot Logic)

If the user reports a failure or hits an impasse at a milestone, the app invokes an **Evaluation Prompt Chain** rather than stalling out completely. It calculates a severity score based on the user's feedback:

Failure Severity Category	Root Cause Example	AI Re-routing Action Taken
Low Severity <i>(Friction)</i>	"I ran out of time this week due to university exams."	Timeline Shift: Automatically recalculates dates, moves Milestone 1 out by 4 days, pushes all downstream calendar dates forward via API, and updates the UI.
High Severity <i>(Structural Roadblock)</i>	"I talked to 10 students, and absolutely none of them are willing to pay for delivery."	The Tactical Pivot: Keeps original timeline boundaries intact but dynamically rewrites the <i>nature</i> of Milestone 1 and 2. It suggests a lower-cost alternative path to explore (e.g., peer-to-peer sharing model).

5. Technical Stack Suggestions (Optimized for 4-Day Build)

To keep implementation fast and clean for a 2-person team, don't build a complex backend from scratch:

- **Frontend Dashboard:** **Streamlit** (Python-based, lightning-fast interactive state components) or **Softr/Bubble** (No-code UI connected to an Airtable database).
- **AI Orchestration Backend:** **LangChain / LangGraph** (Python) or **Flowise** (Visual no-code LLM orchestration tool). This lets you easily chain the Stage 1 inputs directly into Stage 2 questions, and pass those answers into Stage 4.
- **Calendar Integration:** **Make.com** or **Zapier** webhooks. When the app outputs a confirmed roadmap, send a JSON payload to Make.com to easily handle the Google Calendar event creation without writing heavy OAuth code from scratch.

6. Submission Copywriting Blocks (Ready for Devpost)

Here is exactly how you write up the mandatory evaluation responses for your Devpost portal submission:

AI Architecture Explanation

Inputs: Unstructured project goals text from user, human constraint parameters (hours, cash), real-time progress update responses. **AI Capability Used:** Chain-of-Thought Natural Language Processing (NLP), Decision-Support Framework Chains, and Automated Conditional Prompt-Routing. **Processing Engine:** LaunchPad uses a sequential LLM engine. Step 1 parses the input text for underlying assumptions. Step 2 acts as an interrogator evaluating logic gaps. Step 3 processes the validation strategy against Lean Startup frameworks to output a calendar-bound execution plan. Step 4 handles dynamic re-routing when milestone metrics are missed. **Outputs:** An interactive, adaptive project validation dashboard, programmatic Google Calendar milestone notifications, and custom strategic pivot roadmaps.

Human-in-the-Loop Design

The Boundary: Our AI explicitly refuses to make the final "Pivot or Kill" decision on an idea. **The Justification:** While the system structures data, flags structural inconsistencies, and models failure severity risks, it cannot simulate authentic human intuition, local market nuances, or a founder's resilience. The tool acts as a thinking mirror; the ultimate choice to risk capital or shift strategies remains entirely with the human user.

Responsible AI Guardrails

Realistic Risk identified: False Confidence & Over-Reliance. A user might assume an AI-generated milestone framework guarantees a successful business outcome, leading to financial loss or frustration if it fails. **Concrete Mitigation implemented:** We implemented clear Uncertainty Framing across all dashboard generation states. Every milestone generated includes explicit "Hypothesis Testing" language, emphasizing that the plan tracks assumptions to disprove, rather than definitive steps to success. Furthermore, the UI enforces a hard programmatic validation gate before subsequent project phases can be unlocked.

This framework is ironclad and fully responsive to every dimension of the undergrad track judging criteria.